

STRUCTURE OF SKIN

A patch of skin the size of a fingernail contains 5 million cells of at least a dozen main kinds, 100 sweat glands, 1,000 touch sensors, 100-plus hairs with sebaceous glands, up to 1m ($3\frac{1}{3}$ ft) of tiny blood vessels, and about 0.5m ($1\frac{2}{3}$ ft) of nerve fibres.

Touch sensor

Specialized nerve ending at edge of epidermis; other touch sensor types lie deeper in dermis

Hair shaft

Part of hair that projects above skin surface

Arrector pili muscle

Tiny muscle that pulls up hair when cold

Sweat

Drops of perspiration ooze from sweat pores

Basal epidermal layer

Layer in which fast cell division renews epidermis above

Epidermal surface

Cornified layer of flat, dead skin cells

Capillaries

Tiny blood vessels that supply oxygen and nutrients to tissues and collect waste

Epidermis

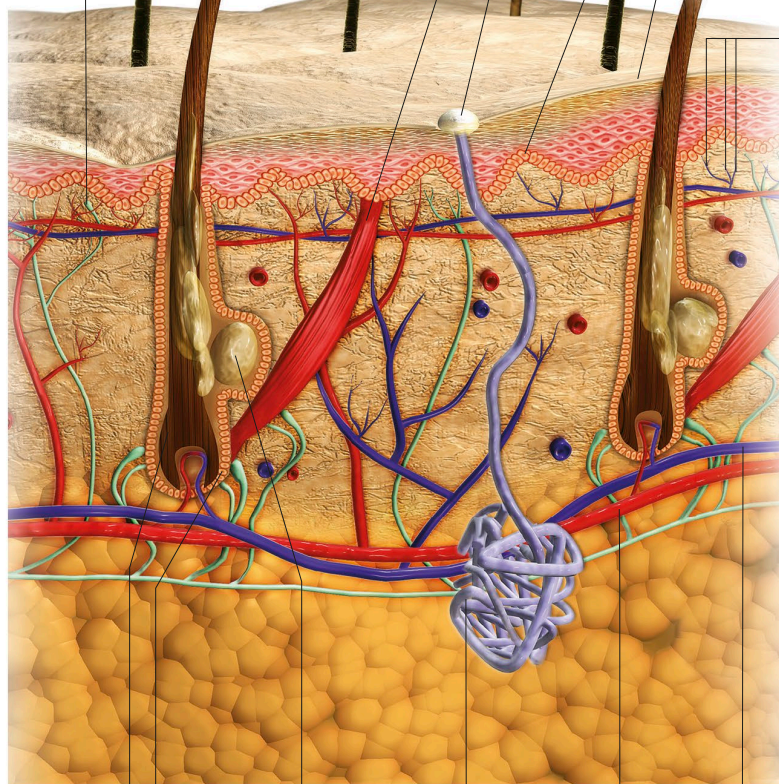
Outer protective layer consisting of tough, flat cells

Dermis

Layer containing blood vessels, glands, and nerve endings

Subcutaneous fat

Acts as an insulator, shock absorber, and energy store

**Hair bulb**

Lowest part of hair, where growth occurs

Hair follicle

Pouch of epidermis at root of hair

Sebaceous gland

Produces sebum that protects hair and lubricates skin

Sweat gland

Coiled knot of tubes secreting watery sweat

Arteriole

Supplies oxygenated blood

Venule

Carries away waste